PROOF SERIES — SHUNYAYA STRUCTURAL DIAGNOSIS (SSD)

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Caution: Research and observation only. Not for critical or automated decision-making.

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0. EXECUTIVE SUMMARY — SSD PROOF SERIES

This document exists for one purpose:

To demonstrate, using **executable and reproducible evidence**, that **Shunyaya Structural Diagnosis (SSD)** is deterministic, conservative, and operational on real mathematical systems.

SSD answers a prior question that classical mathematics and engineering do not formally ask:

"Is this result, system, or process structurally admissible to be relied upon here — even if it is correct?"

SSD does **not** alter mathematics.
SSD does **not** optimize systems.
SSD does **not** predict outcomes.

Instead, SSD **diagnoses**, post-hoc and conservatively, whether **structural stability is eroding** and whether **reliance is becoming unsafe**, while preserving **exact classical correctness at all times**.

This Proof Series is intentionally separate from the main SSD document:

• The **SSD Core Document** defines principles, scope, and boundaries
• The **SSD Proof Series** demonstrates executable, trace-backed diagnostic behavior

No simulations are used.
No stochastic processes are introduced.
No learned or generated training data is used.

All traces are produced by **deterministic execution of classical mathematical procedures**, including explicitly defined corridor evaluations where applicable.
No solver behavior is modified.
No runtime intervention occurs.

No classical mathematics is altered.

Every numerical value collapses **exactly** to its classical counterpart under the invariant:

```
phi((m, a, s)) = m
```

0.1 ROLE OF THIS DOCUMENT IN THE SSD ECOSYSTEM

SSD is published using a **two-document discipline**.

Document 1 — SSD Core Document (Conceptual)
Defines what SSD is, what it diagnoses, and what it deliberately does not do.

Document 2 — SSD Proof Series (This Document)
Demonstrates that SSD operates **deterministically on real traces**, producing meaningful stability diagnostics **without altering results**.

This separation is deliberate.

Principles define responsibility.
Proof establishes credibility.

0.2 WHAT THIS DOCUMENT PROVES — AND WHAT IT DOES NOT

This document proves that:

- SSD detects **structural instability before failure**
- SSD distinguishes **mixed regimes from stable corridors**
- SSD assigns **explainable diagnostic signals**
- SSD remains **silent where structure is admissible**
- SSD preserves **exact classical outputs**

This document does **not** claim:

- improved numerical accuracy
- better convergence
- optimization gains
- predictive power
- governance authority

SSD is a **diagnostic framework**, not a control framework.

0.3 PROOF METHODOLOGY

Each proof case follows a fixed, deterministic structure:

1. A real classical execution is selected
2. Classical computation is run unchanged
3. Structural execution traces are recorded
4. SSD is applied post hoc to those traces
5. Diagnostic outcomes are evaluated

SSD operates strictly on **recorded structural traces**.

Traces are required only to expose classical outputs and **minimal structural signals** (posture, accumulation, repetition), following a **solver-agnostic trace schema**.

All diagnostic outcomes, including **DENY** and **ABSTAIN**, are **derived verdicts** computed post hoc from trace signals.
They do **not** need to exist as stored trace `state` values.

SSD does **not** require:

• solver instrumentation
• runtime hooks
• inline evaluation
• learning or adaptation
• Shunyaya-specific execution engines

No runtime intervention occurs.
No solver behavior is modified.

SSD consumes traces only.
It never influences execution.

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1. CASE 1 — NONLINEAR SOLVER REGIME DIAGNOSIS (MGH17)

1.1 Objective of This Case

To demonstrate that Shunyaya Structural Diagnosis (SSD) can distinguish:

• mixed structural regimes
• unstable reliance patterns
• stable admissible corridors

using only **deterministic diagnostic signals**.

1.2 System Under Observation

• Classical system: Gauss–Newton nonlinear least squares
• Dataset: MGH17 benchmark
• Data points: 33
• Known properties: sensitivity to initialization and conditioning

The solver is executed **exactly as in classical mathematics**.
SSD is applied **only after execution**, using recorded traces.

1.3 Observed Structural Regimes

Three solver runs are evaluated.

Run A — Structural Undefinedness

• Singular normal matrix at iteration 1
• Classical solver halts
• Structural posture undefined

Run B — Mixed Regime

• Classical solver continues numerically
• Structural signals fluctuate
• Accumulated pressure increases
• Structural denial events present

Run C — Stable Corridor

• Solver remains well-conditioned
• Structural posture remains centered
• Accumulation remains bounded
• Consistent admissibility

1.4 SSD Diagnostic Execution

SSD is applied to **recorded structural traces** using a deterministic post-run diagnostic pass.

SSD computes:

• posture drift
• accumulation growth
• regime variance
• **deny_rate and abstain_rate (derived diagnostic verdicts)**

DENY and ABSTAIN are **computed post hoc** from trace signals.
They do **not** need to exist as stored trace `state` values.

No thresholds are learned.
All metrics are **explicit and auditable**.

1.5 Diagnostic Results — Mixed Regime

• Total drift: high
• Stability score `S`: low
• `deny_rate`: non-zero
• `abstain_rate`: non-zero

Interpretation

The system exhibits structural inconsistency.
Reliance would be unsafe **despite classical execution**.

1.6 Diagnostic Results — Stable Corridor

• Total drift: low
• Stability score `S`: approximately `0.85`
• `deny_rate`: `0`
• `abstain_rate`: `0`

Interpretation

The system operates within a **structurally admissible corridor**.
SSD remains silent.

1.7 Why This Case Matters

Most real-world failures occur **after instability has already accumulated**.

• Classical monitoring observes outcomes
• SSD diagnoses erosion

This case demonstrates that:

• stability is detectable **before failure**
• diagnosis does **not** require intervention
• deterministic signals are sufficient

1.8 Evidence (Case 1)

The following figure presents the **canonical structural evidence snapshot** for **Case 1**.

It summarizes the **dominant drift contributors** and the resulting **stability assessment** produced by Shunyaya Structural Diagnosis (SSD), computed deterministically from recorded nonlinear solver traces on the **MGH17 benchmark**.

The figure is **evidence-only**:

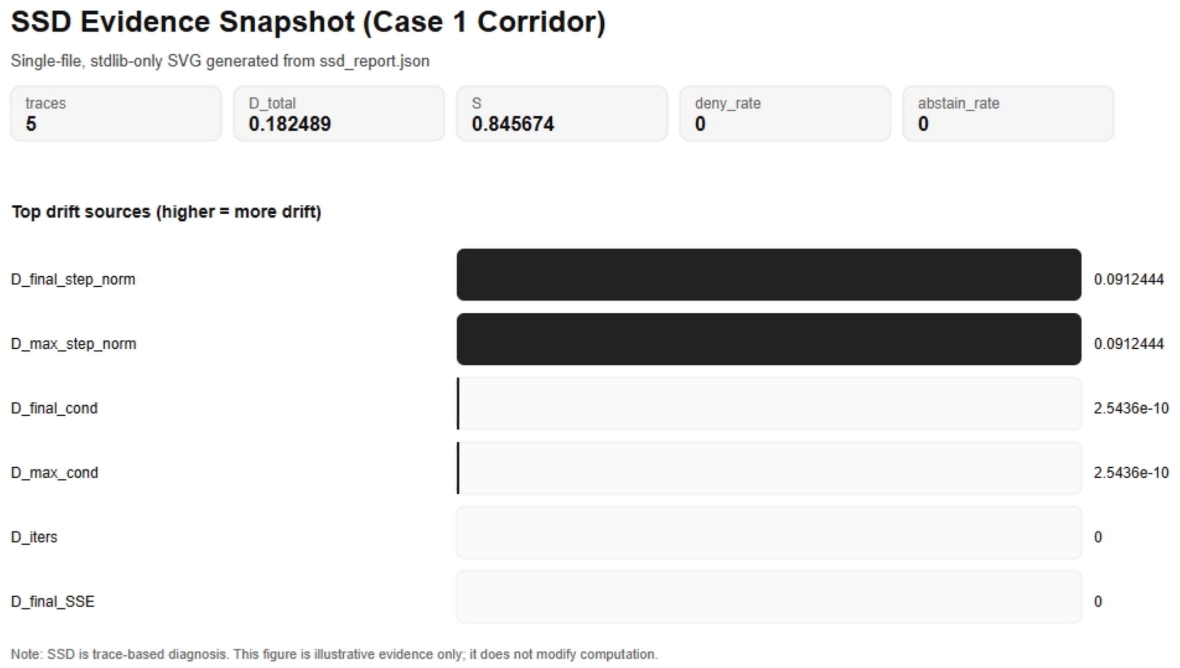
• it reflects **post-execution diagnosis**
• it does **not** modify execution
• it does **not** influence solver behavior
• it does **not** reinterpret the underlying mathematics

The underlying **Gauss–Newton computation remains classical and unchanged**.

Figure — Case 1 (Nonlinear Solver, MGH17) SSD Evidence Snapshot

Canonical diagnostic snapshot showing dominant drift sources and the resulting stability score across **mixed solver regimes**.

Generated deterministically from recorded solver traces using a **conservative, post-hoc structural diagnosis**.



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2. CASE 02A — CALCULUS CORRIDOR DIAGNOSIS (SQRT)

Objective
To demonstrate that Shunyaya Structural Diagnosis (SSD) can reliably diagnose unsafe reliance regions in classical calculus near boundary and undefined zones, without modifying any calculus computation.

This case uses first-order linearization of the function `f(x) = sqrt(x)` under three deterministic corridor regimes.

2.1 Classical Setup (unchanged)

For a fixed step `h`, classical calculus computes:

• True value: `y_true = f(x + h)`
• Linear approximation: `y_lin = f(x) + f'(x) * h`
• Absolute error: `err_abs = |y_true - y_lin|`

All classical quantities are preserved exactly.

2.2 Structural Overlay (diagnostic only)

Post-run structural quantities are derived per step:

• **Risk ratio**:
`r = (|f''(x)| * |h|) / (1 + |f'(x)|)`

• **Permission**:
`a = 1 / (1 + r)`

• **Resistance accumulation**:
`s` accumulates only when `r > r_safe`

• **Diagnostic verdict** in `{ALLOW, DENY, ABSTAIN}`

ALLOW, DENY, and ABSTAIN are **derived diagnostic verdicts**, computed post hoc from trace signals.
They do **not** need to exist as stored trace `state` values.

No value feeds back into the classical computation.
SSD consumes traces only.

2.3 Corridor Definitions

Three deterministic corridors are evaluated:

1. **ALLOW corridor**
   Safe interior region, far from singularity.
2. **DENY corridor**
   Boundary-adjacent region with rapidly increasing curvature.
3. **ABSTAIN corridor**
   Undefined region (`x <= 0`) where calculus is not valid.

Each corridor produces one `trace_sse.csv`, and SSD diagnoses all traces together.

2.4 SSD Diagnostic Results (Case 02A)

• Number of traces: 3
• `D_total = 10.133228660344084`
• Stability score:
`S = 1 / (1 + D_total) = 0.08982120375933193`
• `deny_rate = 0.3333333333333333`
• `abstain_rate = 0.3333333333333333`

2.5 Dominant Drift Sources

The strongest contributors to instability were:

- `D_final_s = 2.9999999999905276`
- `D_max_s = 2.9999999999905276`
- `D_final_err_abs = 1.9990821123918814`
- `D_iters = 0.9620160701240222`
- `D_final_a = 0.08606523892356324`
- `D_min_a = 0.08606523892356324`

2.6 Interpretation

This case demonstrates that:

- Classical calculus may continue to compute values near boundaries
- Structural risk and resistance rise sharply before outright failure
- SSD makes unsafe reliance zones explicit via denial and abstention signals
- **No modification, correction, or optimization of calculus is performed**

SSD therefore changes the question from:

"Can calculus compute?"
to:
"Is it structurally safe to rely on this computation here?"

2.7 Evidence (Case 02A)

The following figure presents the canonical structural evidence snapshot for **Case 02A**.

It summarizes the dominant drift contributors and the resulting stability assessment produced by Shunyaya Structural Diagnosis (SSD), computed deterministically from recorded calculus traces.

The figure is **evidence-only**:

- it reflects post-execution diagnosis
- it does not modify computation
- it does not influence execution
- it does not reinterpret classical calculus

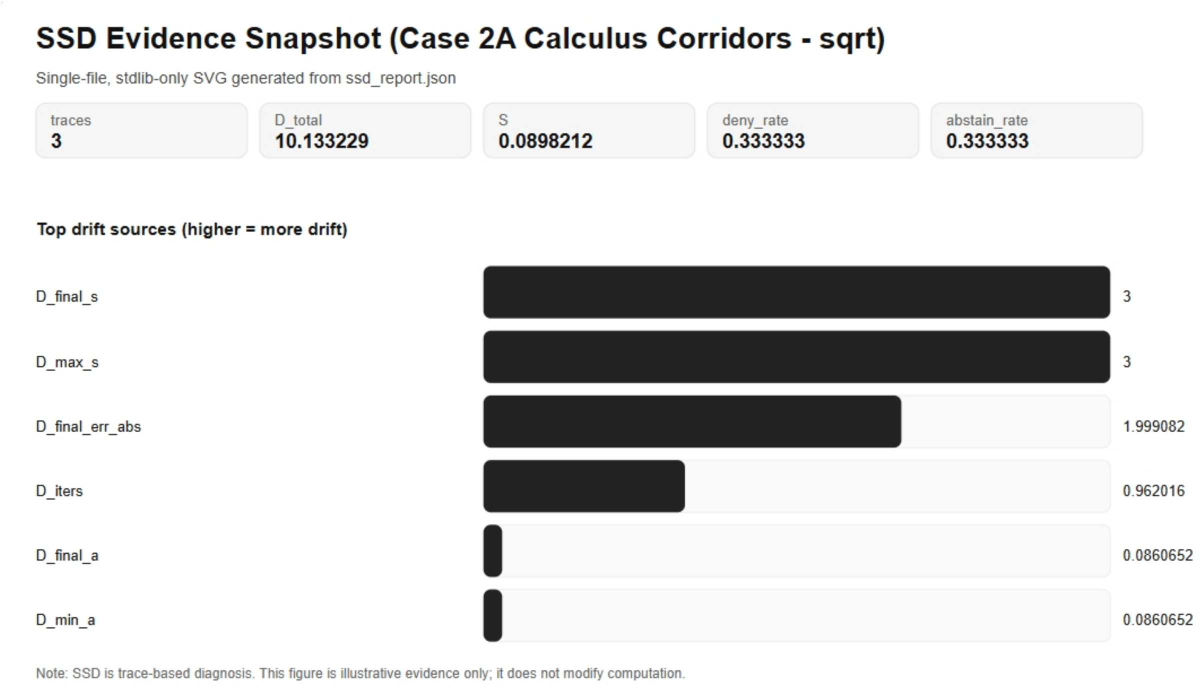

Figure — Case 02A (Calculus, sqrt) SSD Evidence Snapshot
Canonical diagnostic snapshot showing dominant drift sources and the resulting stability score for mixed calculus corridors.
Generated deterministically from recorded traces using a conservative, post-hoc structural diagnosis.

2.2 CASE 02B — CALCULUS CORRIDOR DIAGNOSIS (1 / (1 − x))

2.2.1 Objective

To demonstrate that Shunyaya Structural Diagnosis (SSD) can diagnose unsafe reliance near a classical calculus boundary where the function becomes unstable as `x -> 1`, without modifying any classical computation.

This case uses first-order linearization of the function `f(x) = 1 / (1 - x)` under three deterministic corridor regimes.

2.2.2 Classical Setup (unchanged)

For a fixed step `h`, classical calculus computes:

• True value: `y_true = f(x + h)`
• Linear approximation: `y_lin = f(x) + f'(x) * h`
• Absolute error: `err_abs = |y_true - y_lin|`

All classical quantities are preserved exactly.

2.2.3 Structural Overlay (diagnostic only)

Post-run structural quantities are derived per step:

• **Risk ratio**:
`r = (|f''(x)| * |h|) / (1 + |f'(x)|)`

• **Permission**:
`a = 1 / (1 + r)`

• **Resistance accumulation**:
`s` accumulates only when `r > r_safe`

• **Diagnostic verdict** in `{ALLOW, DENY, ABSTAIN}`

ALLOW, DENY, and ABSTAIN are **derived diagnostic verdicts**, computed post hoc from trace signals.
They do **not** need to exist as stored trace `state` values.

No value feeds back into the classical computation.
SSD consumes traces only.

2.2.4 SSD Diagnostic Results (Case 02B)

• Number of traces: 3
• `D_total = 11.3711`
• Stability score:
`S = 1 / (1 + D_total) = 0.0808336`
• `deny_rate = 0.333333`
• `abstain_rate = 0`

2.2.5 Dominant Drift Sources

The strongest contributors to instability were:

- `D_final_s = 3`
- `D_max_s = 3`
- `D_final_err_abs = 2.99998`
- `D_final_a = 0.649798`
- `D_min_a = 0.649798`
- `D_iters = 0.404844`

2.2.6 Interpretation

This case demonstrates that:

- Classical calculus continues to compute in regions approaching a pole
- Structural resistance and error drift sharply as reliance becomes unsafe near the boundary
- SSD detects this erosion via explicit denial patterns and concentrated drift
- **No modification, correction, or optimization of calculus is performed**

SSD therefore changes the question from:

"Can calculus compute?"
to:
"Is it structurally safe to rely on this computation here?"

2.2.7 Evidence (Case 02B)

The following figure presents the canonical structural evidence snapshot for Case 02B.

It summarizes the dominant drift contributors and the resulting stability assessment produced by Shunyaya Structural Diagnosis (SSD), computed deterministically from recorded calculus traces.

The figure is evidence-only:
- it reflects post-execution diagnosis
- it does not modify computation
- it does not influence execution
- it does not reinterpret classical calculus

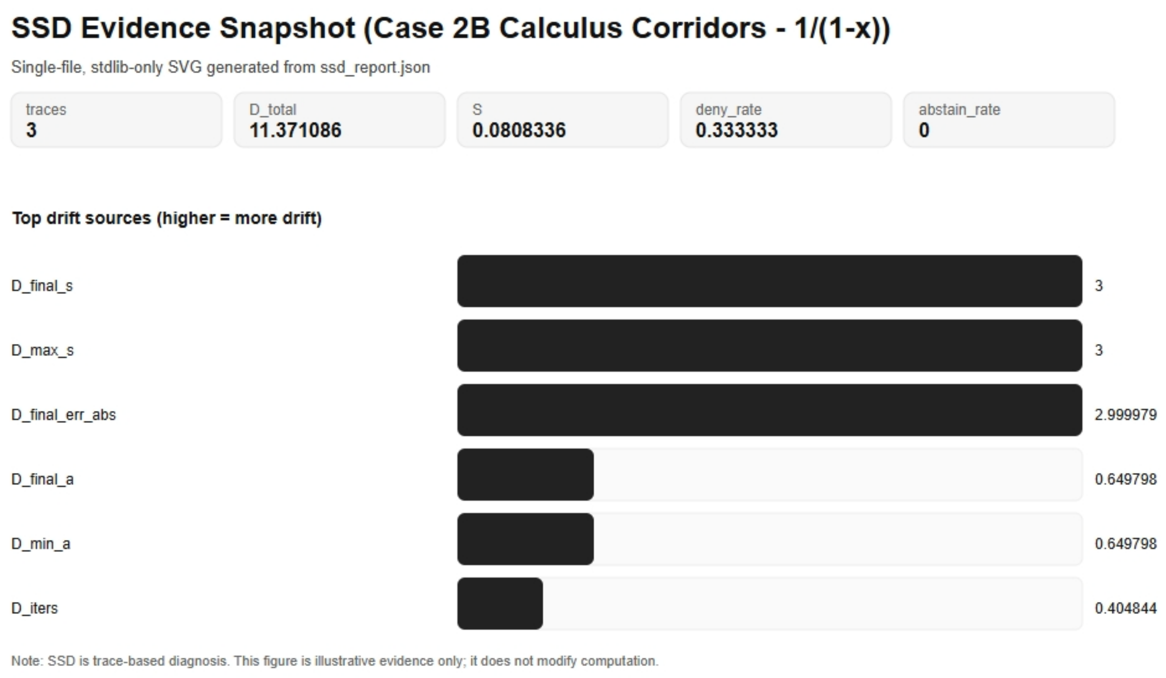

Figure — Case 02B (Calculus, 1/(1 − x)) SSD Evidence Snapshot
Canonical diagnostic snapshot showing dominant drift sources and the resulting stability score for mixed calculus corridors near a boundary pole.
Generated deterministically from recorded traces using a conservative, post-hoc structural diagnosis.

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3. CASE 03 — LINEAR SYSTEM CONDITIONING CORRIDOR

Objective
To demonstrate that Shunyaya Structural Diagnosis (SSD) can diagnose unsafe reliance in linear algebra systems caused by conditioning effects, even when classical solvers continue to compute valid numerical results.

This case evaluates deterministic linear systems under controlled conditioning corridors to expose **stability erosion driven by matrix conditioning**, rather than numerical failure.

3.1 System Under Observation

• Classical system: Linear system solving `A x = b`
• System type: Deterministic square linear system
• Known properties: Sensitivity to matrix conditioning
• Solution method: Classical direct solve (unchanged)

The linear system is solved exactly as in classical linear algebra.
SSD is applied only after execution, using recorded traces.

3.2 Observed Structural Regimes

Three linear system corridors are evaluated.

Run A — Stable Conditioning (ALLOW)
• Well-conditioned matrix
• Classical solution stable
• Structural posture remains centered
• Accumulation remains bounded

Run B — Ill-Conditioned System (DENY)
• Matrix condition number elevated
• Classical solution continues numerically
• Structural risk rises sharply
• Accumulated pressure increases
• Structural denial events present

Run C — Singular / Near-Singular System (ABSTAIN)
• Matrix singular or numerically unstable
• Classical solve becomes undefined or unreliable
• Structural posture undefined
• Abstention signaled without intervention

3.3 SSD Diagnostic Execution

SSD is applied to recorded structural traces using a deterministic post-run diagnostic pass.

SSD computes:

• posture drift
• accumulation growth
• regime variance
• denial and abstention rates

No thresholds are learned.
All metrics are explicit.

3.4 SSD Diagnostic Results (Case 03)

• Number of traces: 3
• `D_total = 18.5471`
• Stability score:
`S = 1 / (1 + D_total) = 0.0511584`
• `deny_rate = 0.333333`
• `abstain_rate = 0.333333`

3.5 Dominant Drift Sources

The strongest contributors to instability were:

• `D_final_s = 3`
• `D_max_s = 3`
• `D_final_cond = 2`
• `D_max_cond = 2`
• `D_final_r = 2`
• `D_final_risk = 2`

3.6 Interpretation

This case demonstrates that:

• Classical linear solvers may continue to produce solutions even under severe conditioning
• Conditioning-induced instability accumulates structurally before numerical failure
• SSD makes unsafe reliance explicit via denial and abstention signals
• **No modification, correction, or optimization of linear algebra is performed**

SSD therefore changes the question from:

“Can the linear system be solved?”
to:
“Is it structurally safe to rely on this solution here?”

3.7 Why This Case Matters

Linear systems underpin nearly all numerical computation.

This case demonstrates that:

• conditioning risk can remain invisible to classical correctness checks
• instability emerges structurally before breakdown
• deterministic diagnosis is possible without altering linear algebra

SSD exposes reliance risk where classical mathematics remains silent.

3.8 Evidence (Case 03)

The following figure presents the canonical structural evidence snapshot for **Case 03**.

It summarizes the dominant drift contributors and the resulting stability assessment produced by Shunyaya Structural Diagnosis (SSD), computed deterministically from recorded linear system traces across conditioning corridors.

The figure is **evidence-only**:

• it reflects post-execution diagnosis
• it does not modify computation
• it does not influence execution
• it does not reinterpret the underlying classical linear system

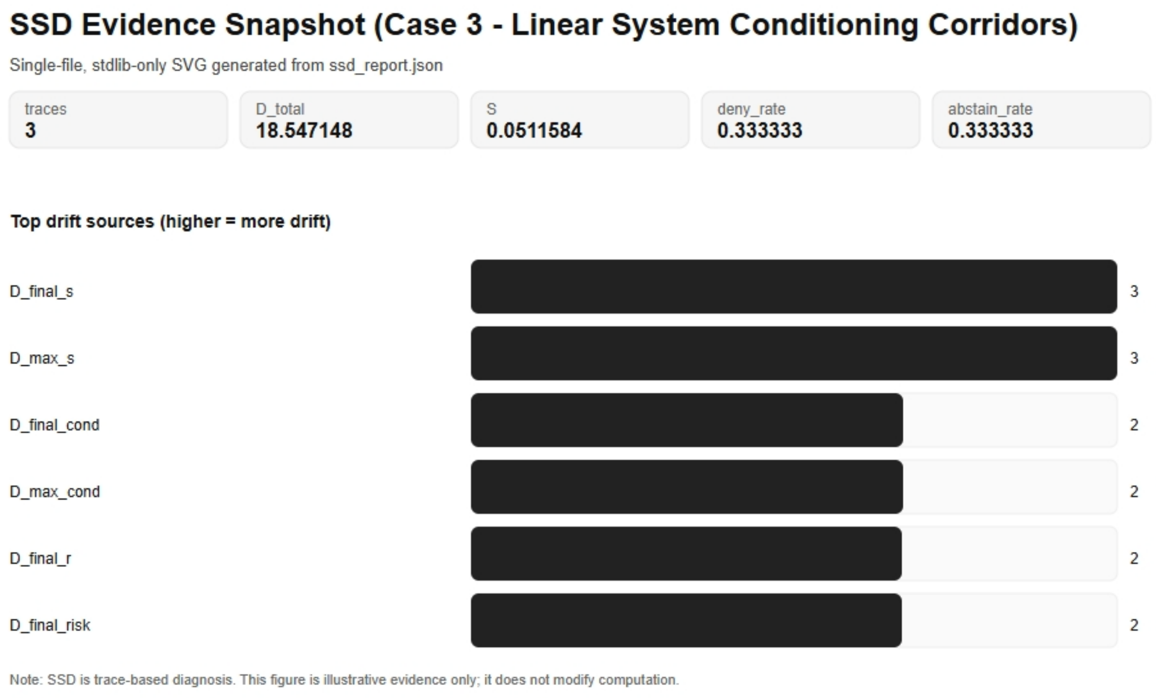

Figure — Case 3 (Linear System Conditioning Corridors) SSD Evidence Snapshot
Canonical diagnostic snapshot showing dominant drift sources and the resulting stability score across stable, ill-conditioned, and singular linear system regimes.
Generated deterministically from recorded traces using a conservative, post-hoc structural diagnosis.

4. CASE 4A — ODE TIME-EVOLUTION EROSION DIAGNOSIS (STIFFNESS / TEMPORAL FATIGUE)

Objective
To demonstrate that Shunyaya Structural Diagnosis (SSD) can diagnose time-evolving structural erosion in dynamical systems — including cases where classical integration continues to compute finite numerical values, while reliance becomes structurally inadmissible due to accumulated fatigue.

This case completes SSD's foundational mathematical coverage by extending diagnosis from **static systems to time evolution**.

4.1 System Under Observation

• Classical system: first-order ODE time evolution
• ODE: `y'(t) = -k * y(t)`, with initial condition `y(0) = 1`

• Classical reference solution (for error measurement only):
`y_true(t) = exp(-k * t)`

• Classical solver used (unchanged): explicit Euler time stepping
`y_{n+1} = y_n + dt * (-k * y_n) = y_n * (1 - k * dt)`

The solver is executed exactly as in classical numerical analysis.
SSD is applied only after execution, using recorded traces.

4.2 Structural Overlay (diagnostic only)

Post-run structural quantities are derived per time step:

• **Amplification factor**:
`g = |1 - k * dt|`

• **Risk scalar**:
`r = (k * dt) * g`

• **Permission**:
`a = 1 / (1 + r)`

• **Resistance (fatigue) accumulation**:
• if `r <= r_safe`, then `s = max(0, s - s_relief)`
• if `r > r_safe`, then `s = s + (r - r_safe)`

• **Diagnostic verdict** in `{ALLOW, DENY, ABSTAIN}`

ALLOW, DENY, and ABSTAIN are **derived diagnostic verdicts**, computed post hoc from trace signals.
They do **not** need to exist as stored trace `state` values.

• **ABSTAIN** when time evolution is structurally divergent
(`g > g_abstain`) or becomes non-finite

• **DENY** when accumulated fatigue crosses admissibility
(`s > s_max`) or permission falls below minimum
(`a < a_min`)

• **ALLOW** otherwise

No value feeds back into the classical computation.
SSD consumes traces only.

4.3 Corridor Definitions

Three deterministic corridors are evaluated.

Run A — ALLOW (Stable Time Evolution)
• Parameters selected so `g < 1` and structural risk remains low
• Resistance remains bounded
• Classical computation proceeds normally
• SSD remains silent

Run B — DENY (Erosion While Still Computing)
• Parameters selected so time evolution remains finite and computable
• Structural fatigue accumulates steadily
• DENY occurs while classical values remain finite
• Reliance becomes unsafe without numerical failure

Run C — ABSTAIN (Divergent / Non-Admissible Time Evolution)
• Parameters selected so `g > g_abstain`
• Structural divergence detected immediately
• SSD abstains without intervention

Each corridor produces one `trace_sse.csv`, and SSD diagnoses all traces together.

4.4 SSD Diagnostic Results (Case 4A)

• Number of traces: 3
• `D_total = 23.3117`
• Stability score:
`S = 1 / (1 + D_total) = 0.0411325`
• `deny_rate = 0.333333`
• `abstain_rate = 0.333333`

4.5 Dominant Drift Sources

The strongest contributors to instability were:

• `D_final_s = 3`
• `D_max_s = 3`
• `D_final_step_norm = 2.99982`
• `D_final_err_abs = 2.99958`
• `D_final_r = 2.5382`
• `D_final_risk = 2.5382`

4.6 Interpretation

This case demonstrates that:

• Dynamical systems may remain numerically computable while becoming structurally inadmissible to rely upon
• Structural fatigue becomes the dominant failure precursor in time-evolving systems
• SSD can deny reliance **before** classical solvers fail, without modifying computation
• SSD can abstain immediately under divergent evolution

SSD therefore changes the question from:

"Can the ODE be integrated?"
to:
"Is it structurally safe to rely on this time evolution here?"

4.7 Why This Case Matters

Real systems fail through accumulation — erosion before breakdown and fatigue before collapse.

Case 4A establishes that SSD detects **time-accumulated instability** deterministically, post-hoc, and without altering classical mathematics.

4.8 Evidence (Case 4A)

The following figure presents the canonical structural evidence snapshot for **Case 4A**.

It summarizes dominant drift contributors and the resulting stability assessment produced by Shunyaya Structural Diagnosis (SSD), computed deterministically from recorded ODE time-evolution traces across ALLOW, DENY, and ABSTAIN corridors.

The figure is **evidence-only**:

• it reflects post-execution diagnosis
• it does not modify computation
• it does not influence execution
• it does not reinterpret the underlying classical ODE computation

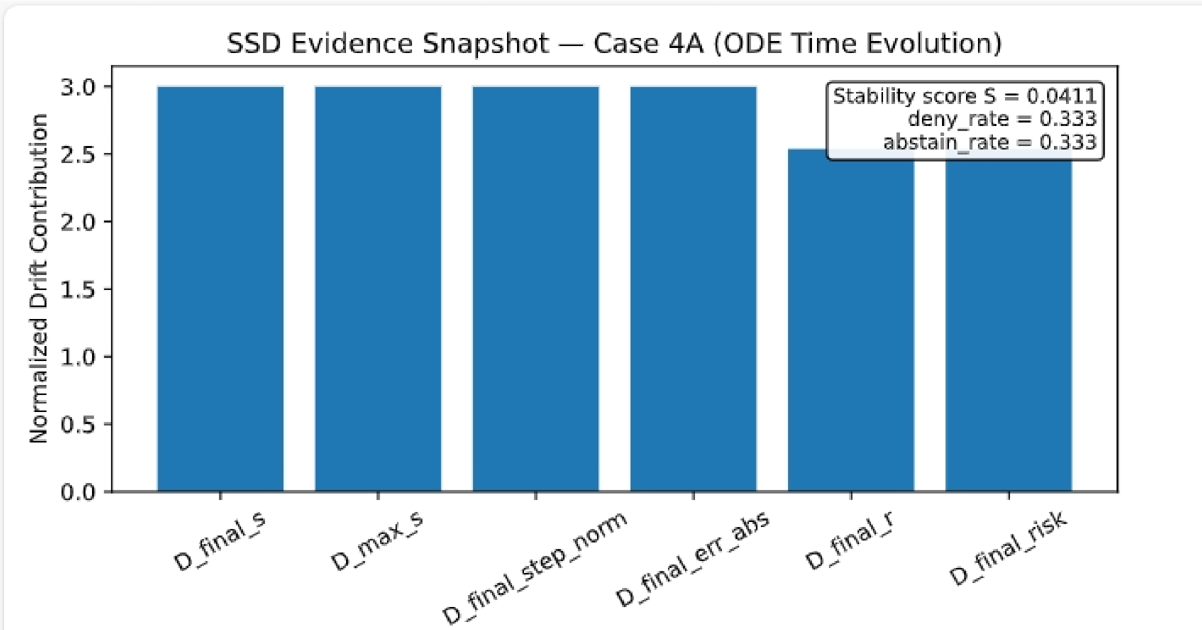

Figure — Case 4A (ODE Time-Evolution Corridors) SSD Evidence Snapshot
Canonical diagnostic snapshot showing dominant drift sources and the resulting stability score across stable evolution, erosion-denial, and divergent abstention regimes. Generated deterministically from recorded traces using a conservative, post-hoc structural diagnosis.

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5. REPRODUCIBILITY STATEMENT

All results in this proof are:

• deterministic
• replayable

• trace-backed
• machine-independent

All diagnostics are derived exclusively from recorded structural traces following the **minimal SSD trace schema**.

Given identical traces, SSD guarantees **identical diagnostic outcomes**, **identical stability scores**, and **identical drift attribution**.

Independent reproduction is expected.

5A. Evidence Artifacts and Visualization Policy

All figures referenced in this proof series are derived exclusively from deterministic SSD diagnostics computed post-execution from recorded traces.

For each case or subcase, SSD may produce one or more **canonical evidence snapshots** in vector format (SVG).
These SVG snapshots are the **authoritative, citation-ready evidence artifacts** used for documentation, review, and verification.

In addition, optional raster or interactive visualizations (for example, Matplotlib-generated PNG plots) may be generated for reference or presentation purposes.
Such plots:

• render only diagnostics already computed from traces
• do not regenerate traces
• do not recompute SSD metrics
• do not reinterpret or normalize results
• do not influence any computation

All diagnostic conclusions in this proof series rely exclusively on canonical evidence artifacts and recorded numeric results.
Optional visualizations are provided for convenience only and do not carry evidentiary authority.

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6. POSITIONING OF THIS PROOF SERIES

This Proof Series establishes **minimum credibility** for Shunyaya Structural Diagnosis (SSD).

It demonstrates **diagnostic value without expanding scope**.

Future proof cases may extend to:

• additional calculus regimes
• system workflows
• dependency graphs

These extensions are **additive**, not required.

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7. CONCLUSION

SSD introduces a new capability:

the ability to diagnose when stability is eroding — **without altering mathematics**.

This document demonstrates that capability in **executable, reproducible form**.

SSD does not control.
SSD does not predict.
SSD diagnoses.

OMP